



BIOLOGY MISCELLANEOUS — FULL MIND MAP

Biology Ch. 11 · Miscellaneous · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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1. Public Health Programmes, Surveys & Schemes

- AYUSH ka full form Ayurveda, Yoga and Naturopathy, Unani, Siddha aur Homeopathy hai; ye India mein practiced six Indian systems of medicine ko represent karta hai. [Q1]
- 'Red Ribbon Express' HIV/AIDS ke transmission, prevention aur risk-perception ke baare mein awareness create karne wala moving visual medium hai. [Q2]
- SANRAKSHA (Bengaluru) 1993 mein HIV counselling service ke roop mein start hua; source ise AIDS Research and Control Project se link karta hai aur HIV testing se hospice care tak services batata hai. [Q3]
- AVAHAN Bill & Melinda Gates Foundation ki initiative hai jo India mein HIV spread reduce karne ke liye 2003 mein start hui; NACO 1992 mein establish hua aur HIV/AIDS control programme ko lead karta hai. [Q4, Q5]
- Mission Indradhanush 25 Dec 2014 ko launch hua; under-2 children aur pregnant women ko seven vaccine-preventable diseases ke against immunize/vaccinate karna iska objective hai. [Q6, Q7, Q8]
- Anaemia Mukta Bharat (AMB) mein Iron + Folic Acid supplementation, delayed cord clamping, periodic deworming aur endemic pockets mein malaria/haemoglobinopathies/fluorosis jaise non-nutritional causes ko address karna include hai; calcium supplementation wala statement source mein incorrect hai. [Q9]
- National Sickle Cell Anaemia Elimination Mission 1 July 2023 ko Shahdol (M.P.) mein launch hua; source target year 2047 deta hai, especially tribal populations mein sickle-cell genetic transmission eliminate karne ke context mein. [Q10]
- Source ka programme-year matching: National Programme for Control of Blindness → 1976; ICDS → 1975; National Leprosy Eradication Programme → 1983; National Rural Health Mission → 2005. [Q11]
- NFHS-5 mein pre-school education, death registration aur hypertension/diabetes prevalence jaise indicators included the, lekin source ke according HIV test include nahi tha. [Q12]

2. Chemistry, Environment, Microbes & Weeds

- Correct chemistry matching: Chlorophyll → Magnesium; Haemoglobin → Iron; Table salt → Sodium chloride; Washing soda → Sodium carbonate. Isliye 'Washing soda–Potassium' wrong pair hai. [Q13]
- Vinegar mein Acetic acid, sour milk mein Lactic acid, bones mein Calcium phosphate aur gastric juice mein Hydrochloric acid hota hai; fruit ripening ko Ethylene se link kiya gaya hai. [Q14, Q15]
- Environment matching: Skin cancer → Ultraviolet rays; Noise pollution → Decibel; Global warming → Carbon dioxide; Ozone hole → CFCs. [Q16]
- Wool biodegradable material hai aur keratin protein se bana hota hai. Biodegradation mein microorganisms—especially bacteria—organic waste ko simpler/reusable substances mein break karte hain; given options mein insecticides non-biodegradable hain. [Q17, Q18, Q19]

- Microbe-use matching: Lactobacillus → curd; Rhizobium → nitrogen fixation; Yeast → alcohol; Aspergillus → citric acid. Source bacteria, fungi aur microalgae—teeno ko industrially useful batata hai. [Q20, Q21]
- Methanogens archaeobacteria ka group hai jo low-oxygen conditions mein methane produce karta hai. [Q22]
- Saccharomyces yeast toddy fermentation se linked hai; commercial ethanol ke liye Saccharomyces cerevisiae important hai. Yeast baking aur alcoholic beverages dono mein use hota hai. [Q23, Q24, Q25]
- Thermophiles high-temperature loving microorganisms hain; source 50–80°C optimum range aur hyperthermophiles ke liye ~80–110°C tak growth mention karta hai. [Q26]
- Biomagnification ka matlab food chain ke successive trophic levels par pesticides/toxic chemicals ki concentration ka increase hona hai. [Q27]
- Herbicides unwanted weeds ko kill karte hain; selective herbicides target weeds ko damage karte hue desired crop ko relatively safe chhod sakte hain. Source assertion-reason mein A aur R dono true aur R correct explanation hai. [Q28]
- Orobanchae ek parasitic weed hai jo commonly tobacco fields se associated hai; tomato aur potato fields mein bhi mil sakta hai. [Q29]

3. Scientific Days, Awareness & Biodiversity

- Herbarium preserved plant specimens ka systematically stored/catalogued collection hota hai; plant identification aur reference ke liye use hota hai. [Q30]
- ICAR har saal 16 July ko Foundation Day celebrate karta hai; source establishment date 16 July 1929 batata hai. [Q31]
- Important dates: National Maritime Day → 5 Apr; World Health Day → 7 Apr; World Athletics Day → 7 May; World Red Cross Day → 8 May. Source ka chronological order 2→1→3→4 hai. [Q32]
- Other dates: World Environment Day → 5 Jun; World AIDS Day → 1 Dec; Teachers' Day → 5 Sep; World Food Day → 16 Oct; World Homeopathy Day → 10 Apr; Earth Day → 22 Apr; International Yoga Day → 21 Jun. [Q33, Q34]
- World TB Day → 24 Mar; World AIDS Day → 1 Dec; Anti-Leprosy Day → last Sunday of January; International Ozone Day → 16 Sep. Bangkok mein International AIDS Conference 16 July 2004 ko hua tha. [Q35, Q36, Q37, Q38, Q39]

EXAM TRAP — Important Days — Source Matching Issue

► World Water Day 22 March hota hai, 22 April nahi; World Leprosy Day last Sunday of January hota hai, 30 June nahi. Isliye source Q40 mein B aur C dono pairs incorrect batata hai. [Q40]

- World Diabetes Day har saal 14 November ko observe hota hai. [Q41]

EXAM TRAP — World Minority Rights Day — Code Mismatch

► Correct dates: World Minority Rights Day→18 Dec; World AIDS Day→1 Dec; World Animal Day→4 Oct; World Food Day→16 Oct. Source Q42 mein exact correct code options mein nahi tha, isliye answer marked (*) hai. [Q42]

- Animal facts source: Cuckoo apna nest nahi banata aur doosre birds ke nest mein eggs lay karta hai; lizard tail regenerate kar sakti hai; Blue whale largest living creature hai; Chimpanzee ko most intelligent animal diya gaya hai. [Q43]
- Tamil Nadu ne Oct 2022 mein State-Level Committee for Vulture Conservation launch ki; Asian Giant Tortoises Dec 2022 mein Nagaland ke Ntangki National Park mein release kiye gaye. [Q44, Q45]
- IUCN context mein Gharial critically endangered aur Indian Elephant endangered category mein diya gaya hai. Cheetah India mein extinct ho gaya tha aur Project Cheetah ke through reintroduction kiya gaya. [Q46, Q47]

- Source sixth mass extinction ka major cause human activities—over-exploitation, habitat loss/fragmentation, ecosystem destruction, pollution aur climate change—batata hai. [Q48]
- Biomass living organisms se derived biological material hai; 'inbuilt time-keeping system' actually biological clock/circadian rhythm hota hai, biomass nahi. [Q49]

4. Medical Diagnostics, Imaging & Therapy

- Endoscope digestive tract ko examine karne wala light + camera equipped flexible/rigid instrument hai; source ise simple 'camera type' category mein nahi rakhta. [Q50, Q51]
- CT/CAT scan multiple X-ray images ko computer-process karke cross-sectional images banata hai. Stomach X-ray se pehle Barium diya ja sakta hai kyunki wo X-rays absorb karke contrast improve karta hai. [Q52, Q53]
- Diagnostic matching: EEG → Brain; ECG → Heart; EOG → Eyes/retina; EMG → Muscle. EEG brain ki electrical activity record karta hai; Audiogram → Ear aur Mammogram → Breast. [Q54, Q55, Q56, Q57, Q58]
- MRI = Magnetic Resonance Imaging; nuclear magnetic resonance principle par based hai aur strong magnetic fields + radio waves use karta hai. Source clear karta hai ki X-rays aur Gamma rays MRI mein use nahi hote. [Q59, Q60, Q61, Q62]
- BMD = Bone Mineral Density test; osteoporosis diagnose/assess karne aur bone minerals/density measure karne ke liye use hota hai. [Q63, Q64, Q65]
- Lithotripsy ultrasound shockwaves se kidney/bladder/ureter stones ko small pieces mein break karti hai, jo baad mein urine se pass ho sakte hain. [Q66]
- Electron beam radiation therapy kuch cancers—source mein skin lymphoma—ke treatment mein use hoti hai. [Q67]
- Tetraplegia/Quadriplegia ka matlab both hands aur both legs mein total ya partial paralysis hai. [Q68]

5. Metabolism, Fibres, Livestock & Applied Biology

- Nutrition, respiration aur synthesis metabolism ke major aspects hain; metabolism mein catabolism (breakdown for energy) aur anabolism (synthesis) dono include hote hain. [Q69]
- Artificial silk ko Rayon kehte hain; rayon regenerated cellulose fibre hai, modified starch nahi. Oxytocin→Hormone, Aspartame→synthetic sweetener, Neoprene→specialized rubber; Wool→protein, Rubber→natural polymer, Fullerene→carbon allotrope. [Q70, Q71, Q72]
- Silk fibre, hair, nails aur claws keratin protein se related hain; Cotton mainly cellulose hai, protein nahi. Sunn hemp ka fibre stem se milta hai. [Q73, Q74, Q75]
- Cattle artificial insemination ke liye bull semen/sperms Liquid Nitrogen mein store kiye jaate hain. [Q76, Q77]
- Kadaknath Indian chicken/fowl breed hai; Holstein high milk-yielding cattle breed hai. Bhadawari→buffalo, Sindhi/Tharparkar→cow aur Jamunapari→goat—so 'Sheep–Tharparkar' wrong pair hai. [Q78, Q79, Q80]
- Rashtriya Gokul Mission ka focus indigenous cow breeds ko conserve aur develop karna hai. [Q81]
- Useful matching: Myoglobin→muscle cell; Sarpagandha→tranquilizer; Carcinoma→radiotherapy; Haemoglobin→oxygen transport. Bulimia→eating disorder; Cholesterol→egg-yolk; Atropine→alkaloid; Insulin→pancreas. [Q82, Q83]
- Indian scientists ne microorganism-derived cadmium sulphide semiconductor nanoparticle use karke electronic diode fabricate kiya; source yeast Schizosaccharomyces pombe ko link karta hai. [Q84]

6. Medical Substances, Contraception & Scientific Fields

- Anaesthetics sensation/awareness reduce karne ke liye use hote hain; source Chloroform, Nitrous oxide, pentothal sodium etc. mention karta hai. Hospitals mein Nitrous oxide mild anaesthetic gas ke roop mein use hoti hai. [Q85, Q86]
- Specialist matching: Cardiologist→Heart; Nephrologist→Kidney; Urologist→Urinary tract; Oculist→Eye. Neurology nervous system disorders se related branch hai. [Q87, Q88]
- Source IOC context mein Colostrum ko safe dietary supplement batata hai; HGH, Testosterone aur Erythropoietin banned performance-enhancing substances hain. Saccharin artificial sweetener hai, carbohydrate nahi; Phenolphthalein acid-base indicator/laxative hai, antiseptic nahi. [Q89, Q90, Q91]
- Saheli ek non-steroidal oral contraceptive pill hai; CDRI Lucknow ne centchroman/ormeloxifene molecule develop kiya aur source ise once-per-week indigenous pill batata hai. [Q92]
- Foetal growth monitor karne ke liye Ultrasonics/Ultrasound use hota hai. [Q93]
- Scientist-field matching: Birbal Sahni→Palaeobotany; K.C. Mehta→Plant Pathology; P. Maheshwari→Embryology; M.S. Swaminathan→Green Revolution. [Q94]
- Other matching: Verghese Kurien→Co-operative Movement; Malcolm Adiseshiah→Development Economics; A.P.J. Abdul Kalam→Space; Pramod Karan Sethi→Jaipur Prosthetic Foot. [Q95]
- Global Invasive Species Database (GISD) alien/invasive species ki information deta hai aur IUCN ke Invasive Species Specialist Group se associated hai. [Q96]

7. Research Institutes, Gene Banks & Organisations

- IVRI Izatnagar, Bareilly mein located hai; CDRI Lucknow mein hai. [Q97, Q98, Q99]
- Survey of India national mapping/surveying agency hai, Ministry of Science & Technology ke under kaam karti hai; iska headquarters Dehradun mein hai. Central Electrochemical Research Institute Karaikudi (Tamil Nadu) mein hai, Kolkata mein nahi. [Q100, Q101, Q102]
- CIRG Makhdoom near Mathura mein; High Security Animal Disease Laboratory Bhopal mein; CIMAP Lucknow mein; BSIP/Birbal Sahni Institute of Palaeosciences Lucknow mein autonomous/non-CSIR institute hai. [Q103, Q104, Q105, Q106]
- Indian Institute of Vegetable Research Varanasi mein hai, Gorakhpur mein nahi; National Institute for the Mentally Handicapped (now NIEPID) Secunderabad/Hyderabad mein hai. [Q107, Q108]
- Institute-location matching: CDRI→Lucknow; JALMA Institute→Agra; MNNIT→Allahabad/Prayagraj; National Sugar Institute→Kanpur. CIMAP→Lucknow; CDFD→Hyderabad; IMTECH→Chandigarh; NII→New Delhi. [Q109, Q110]
- National Institute of Virology Pune mein; CDFD Hyderabad mein; Indian Vegetable Research Institute Varanasi mein; CISH Lucknow mein (Srinagar nahi); ALIMCO head office Kanpur mein. [Q111, Q112, Q113, Q114, Q115]
- International Rice Genebank IRRI, Los Baños (Philippines) mein hai; International Potato Centre (CIP) Lima, Peru mein; FAO HQ→Rome, Central Agmark Lab→Nagpur, Central Rice Research Institute→Cuttack, Central Potato Research Institute→Shimla. [Q116, Q117, Q118]
- NBRI Lucknow mein hai, Kolkata mein nahi; NBPGR headquarters New Delhi mein; Rajiv Gandhi Centre for Biotechnology Thiruvananthapuram mein hai. [Q119, Q120, Q121]
- World's first camel hospital source ke according Dubai mein hai. Camel hump fat store karta hai; UN ne 2024 ko International Year of Camelids designate kiya. [Q122, Q123, Q124]
- Matching: Testosterone→Hormone; Codeine→Sedative drug; Caoutchouc→rubber-source substance; Eugenol→clove aromatic oil. Anthrax→biowarfare toxin; Thalassemia→defective-gene disease; Surrogacy→womb-renting; Transgenics→genome alteration. [Q125, Q126]
- Quinine malaria treatment ka alkaloid hai, antibiotic nahi. [Q127]

- Source ke according first 'dinosaurium' Hyderabad ke B.M. Birla Science Centre mein khula; WHO headquarters Geneva mein hai; India ka first Dental College 1920 mein Kolkata mein R. Ahmed Dental College tha. [Q128, Q129, Q130]

8. Scientists, Bees, Human Responses & Vaccination

- Linus Pauling ne cancer aur ascorbic acid (Vitamin C) ke field mein research ki; source unke Chemistry Nobel 1954 aur Peace Nobel 1962 ka mention karta hai. [Q131]
- Honey bee landmine detection mein useful social insect hai. Hive members smell se ek-doesre ko identify karte hain; food-source direction/distance communicate karne ke liye waggle dance use hota hai. Karl von Frisch ne honey-bee language/waggle dance decode karne par work kiya. [Q132, Q133, Q134, Q135, Q136]
- Alcohol, marijuana aur opium depressants diye gaye hain, jabki cigarette smoke stimulant hai. Silica gel moisture absorb karne wala desiccant hai. Shanti Swarup Bhatnagar science/CSIR research leadership se associated hain. [Q137, Q138, Q139]
- Eye donation/transplant mein Cornea use hota hai. High humidity mein sweat properly evaporate nahi hota, isliye discomfort hota hai; sweat evaporation endothermic process hai aur body ko cool karta hai. Dehydration mein water ke saath salts—mainly sodium chloride—lose hote hain; sea water dehydration overcome karne ke liye suitable nahi hai. [Q140, Q141, Q142, Q143, Q144]

⚠ EXAM TRAP — BAC — Source Answer Disputed

► BAC = Blood Alcohol Concentration. Source explanation ke according $>0.45\%$ BAC fatal ho sakta hai, jabki biologically $>0.4\%$ bhi fatal ho sakta hai; JKPSC key aur explanation ke beech issue ko source disputed mark karta hai. [Q145]

- Ivan Pavlov Experimental Psychology/classical conditioning se linked hain. Women/children ki voice pitch men se generally higher hoti hai. Source climate-change indicators mein plant fossils ko floral indicator batata hai aur 'long-term change' ko given option-set mein indicator nahi maanta. [Q146, Q147, Q148]
- Cell-function matching: Photosynthesis→Chloroplast; Mineral uptake→Plasma membrane; Respiration→Mitochondria; Protein synthesis→Ribosomes. J.C. Bose plants mein life-response research se associated hain. [Q149, Q150]
- Low temperature food spoilage microorganisms ki activity ko retard karta hai; source Pig ko given animals mein fibre properly digest na karne wala batata hai, jabki goat/cow/elephant ruminant context mein diye gaye hain. [Q151, Q152]
- Philatelist postage stamps collect karta hai; Biojector painless needle-free injector system hai; Robert Gordon Webster avian influenza/influenza-virus expert hain. [Q153, Q154, Q155]
- Edward Jenner smallpox vaccine aur 'Father of Immunology' se linked hain; Jonas Salk ne successful Polio vaccine develop ki. [Q156, Q157, Q158]
- Scientist-discovery matching: Arber & Smith→restriction endonucleases; Feldmann→Agrobacterium T-DNA transgenic plants; Mullis→PCR; Temin & Baltimore→reverse transcriptase. [Q159]
- Francis Collins→Human Genome Project/gene discoveries; Sergey Brin→Google; Jimmy Wales→Wikipedia—source teeno pairs ko correct maanta hai. [Q160]

9. Dating, Soil, Ecology, Plant Technology & Fibres

- William Morris Davis erosion cycle/geographical cycle of erosion concept se linked hain; Auxanometer plant growth-rate measure karta hai; Mestizo Latin America mein mixed European + Indigenous ancestry ke person ke liye term hai. [Q161, Q162, Q163]
- Ancient rocks ki age ke liye Uranium–Lead process use ho sakta hai; organic fossils ki age ke liye Carbon-14 radiocarbon dating; tree ki age annual/growth rings count karke estimate ki ja sakti hai. [Q164, Q165, Q166]

- Clay soil fine particles ki wajah se highest water retention dikha sakti hai; alkaline soils ko gypsum addition, leaching/drainage etc. se reclaim/improve kiya ja sakta hai. [Q167, Q168]
- Butterflies pollination aur crop production mein important hain. Eichhornia/Water hyacinth ko 'Terror of Bengal' kaha jata hai; Red tide algal/dinoflagellate bloom se related phenomenon hai. [Q169, Q170, Q171]
- Spirogyra mainly freshwater green algae hai, ocean mein nahi; source Ulothrix ko fresh + marine water se link karta hai. Mycorrhiza fungus aur vascular plant roots ka symbiotic association hai jo absorption/drought tolerance etc. improve kar sakta hai. [Q172, Q173]
- Biorock technology damaged coral reefs/marine ecosystems restoration se related hai. Phytotron controlled-condition plant research facility hai; Polytunnel/Polyhouse plants grow/protect karne ke liye use hota hai. [Q174, Q175, Q176]
- Fibre matching: Linen→Flax; Coir→Coconut; Mohair→Angora goat; Down→duck/geese feathers. Angora wool/fibre Angora rabbit se milta hai, mohair se alag hai. [Q177, Q178, Q179, Q180]
- Bioluminescence living organism ke andar chemical reaction se produced light hai; source Luciferin + Luciferase/Photoprotein ko key chemicals batata hai. [Q181]

10. Agriculture, Immunity, Policy & General Science

- Green Revolution mein high-yielding crops ke liye more fertilizer aur more water ki need source mein highlight hai. Revolution matching: Green→Food grains; White→Milk; Yellow→Oilseeds; Blue→Fish/Aquaculture. [Q182, Q183]
- Green Revolution ka principal crop source mein Mexican Wheat diya gaya hai; rice nursery ka Dapog method Philippines se aaya; India ki first dwarf rice variety source ke according Jaya thi. [Q184, Q185, Q186]
- Gambusia insectivorous fish ponds/wells mein mosquito larvae/eggs eat karke mosquito population control mein help karti hai. [Q187, Q188]
- Active acquired immunity antibodies/immunoglobulins ki production se develop hoti hai; B-lymphocytes antibodies banate hain. [Q189]
- Science & Technology Policy 2003 ko 3 Jan 2003, 90th Indian Science Congress (Bengaluru) context mein launch kiya gaya; source innovation, resource utilisation, hazard-management technologies aur intellectual property management jaise objectives deta hai. [Q190]
- Source ke according tobacco/products ki sale par ban lagane wala first country Bhutan tha (Dec 2004). Given planets mein Mars life-existence ke liye possible/suitable option diya gaya hai. Natural-system development ko self-sustained process of systems se link kiya gaya hai. [Q191, Q192, Q193]

EXAM TRAP — TIVA / TIBA — Source Ambiguity

► TIVA = Total Intravenous Anaesthesia. Source Q194 mein Chandra Shekhar Azad University wale substance ka context clear nahi hai aur possible intended term TIBA (2,3,5-Triiodobenzoic acid; anti-auxin) ho sakta hai, isliye question marked (*) hai. [Q194]

- Knowledge ko source 'justified true belief' (JTB) se link karta hai. Scientific investigation observation, experimentation aur analysis par based hoti hai aur honesty scientific process ke liye essential hai. Semantic memory general facts/ideas/concepts store karti hai; information ko organize/understand/communicate karne wali mental activity Thinking hai. [Q195, Q196, Q197, Q198]